

Brake Aid User Guide

Thank you for being part of Brake Aid's beta testing program.

Brake Aid provides players of Sim racing games such as *iRacing* and *Assetto Corsa*, a mechanism for fine tuning their braking performance, technique and timing, with the ultimate aim of reducing the player's lap times, as a result of incrementally tweaking their braking technique to get maximum performance from the simulated vehicle's braking system.

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Overview of Brake Aid

Brake Aid is an intelligent real-time driver assistance system designed to improve braking consistency, confidence and precision in supported sim racing titles.

Unlike traditional driving aids that rely on fixed braking references or predefined ideal racing lines, Brake Aid provides personalised, dynamic brake markers that are calculated in real time from a braking dataset specific to the game, track and vehicle currently being driven.

Rather than simply indicating a fixed point on the circuit where braking should begin, Brake Aid continuously calculates when braking should start based on the vehicle's current speed, observed braking characteristics and driver reaction time. This allows the braking guidance to remain valid even when approach speed changes from lap to lap.

Brake Aid has been designed around the concept of **braking datasets**.

A braking dataset contains a complete set of dynamic braking references for every significant braking zone around a particular game, track and vehicle combination. These datasets can either be:

- created automatically by Brake Aid while analysing your own driving,
- resumed from one of your previous driving sessions, or
- imported from another Brake Aid user who has already developed and refined the braking dataset.

This allows new users to begin driving immediately using braking references developed by experienced drivers, while still retaining the ability to personalise individual braking markers to suit their own driving style.

Brake Aid supports two complementary operating modes.

Adaptive Mode continuously analyses your braking performance and automatically generates updated braking datasets as your driving evolves. This mode is ideal for learning new circuits, refining braking technique, or developing braking datasets that can later be shared with other Brake Aid users.

Dataset Hold Mode preserves the current braking dataset indefinitely, allowing drivers to repeatedly practise using the same braking references while making deliberate corner-by-corner adjustments using Brake Aid's offset controls. This mode is particularly suited to imported datasets created by experienced drivers, providing a stable foundation from which drivers can gradually personalise the braking references without affecting the underlying dataset.

At any time during a driving session, the driver may freely switch between Adaptive Mode and Dataset Hold Mode. This flexibility allows Brake Aid to adapt naturally to changing circumstances. For example, a driver may use a shared braking dataset as the basis for an entire race session, temporarily switch into Adaptive Mode to incorporate a newly developed braking zone or changing vehicle performance into the dataset, then return to Dataset Hold Mode to continue refining the updated braking references.

Brake Aid is therefore much more than a collection of dynamic brake markers. It is a complete braking assistance platform that enables braking knowledge to be developed, shared, preserved and progressively refined while remaining personalised to each driver's individual preferences.

Understanding Dynamic Brake Markers

Traditional Brake Markers

Most drivers learn a race circuit by selecting fixed visual references located alongside the track, such as braking boards, marshal posts, bridge supports or changes in the track surface.

These reference points provide a convenient indication of where braking should begin under a particular set of conditions. While effective, they are inherently static. They do not account for changes in vehicle approach speed, driver reaction time, braking technique or vehicle performance. As a result, the driver must continually estimate how these changing conditions should influence the point at which braking begins.

Brake Aid approaches this problem differently.

Rather than asking the driver to judge *where* to brake, Brake Aid continuously calculates *when* braking should begin based on the current state of the vehicle and the braking characteristics contained within the active braking dataset.

Dynamic Brake Markers

The braking references generated by Brake Aid are known as **dynamic brake markers**.

Unlike conventional braking markers, a dynamic brake marker **is not** a fixed geographic location on the race circuit.

Instead, every dynamic brake marker is continuously calculated in real time using information such as:

- Current vehicle speed.
- Vehicle deceleration characteristics.
- Driver reaction time.
- The braking characteristics stored within the active braking dataset.
- The desired target speed and position within the corner.

This allows Brake Aid to determine the precise moment at which braking should begin to reproduce the braking performance described by the current braking dataset.

Because the calculation is performed continuously, the position at which the brake marker is displayed naturally changes as vehicle speed changes.

For example, if the vehicle approaches a corner significantly faster than during the previous lap, Brake Aid will automatically display the brake marker earlier to compensate for the additional braking distance required.

Likewise, if the vehicle approaches the same corner more slowly, the brake marker will be displayed correspondingly later.

The objective is not to reproduce the same braking point on the track.

The objective is to reproduce the same braking outcome.

Reproducing Braking Performance

When Brake Aid signals a dynamic brake marker, it is indicating that if the driver begins braking at that instant, using approximately the same braking technique represented by the active braking dataset, the vehicle should achieve a similar braking result.

This generally means arriving at approximately the same minimum corner speed at approximately the same location within the corner, even if the approach speed to the corner differs from previous laps.

This distinction is fundamental to understanding how Brake Aid operates.

Traditional braking references are fixed positions on the circuit.

Brake Aid's dynamic brake markers represent calculated braking opportunities that continually adapt to changing conditions while preserving consistent corner entry behaviour.

The Active Braking Dataset

Every dynamic brake marker displayed by Brake Aid originates from the **active braking dataset**.

The active dataset represents the complete collection of braking information currently being used by Brake Aid for the selected game, track and vehicle combination.

At any given time, the active dataset may have originated from one of three sources:

- An imported dataset created and shared by another Brake Aid user.
- A dataset previously generated and saved under your own user profile.
- A new dataset currently being developed using Adaptive Mode.

Regardless of its origin, Brake Aid always uses the active dataset to calculate the dynamic brake markers presented during a driving session.

This means the driver experiences exactly the same style of braking guidance whether the dataset was created personally or supplied by another user.

Personalising Brake Markers

Although a braking dataset provides the foundation for Brake Aid's braking guidance, it is not intended to be immutable.

Every dynamic brake marker within the dataset can be individually refined to better suit the driver's own braking style and level of experience.

Brake Aid provides dedicated adjustment controls that allow each corner to be shifted forwards or backwards in precise two-metre* increments.

**Increment distance is customisable within Brake Aid's CONFIGURATION settings page.*

These adjustments allow the driver to progressively personalise the braking dataset while preserving the overall braking strategy represented by the original dataset.

Whether the dataset was imported from an experienced driver or created from the driver's own performance, Brake Aid is designed to encourage gradual refinement rather than forcing the driver to estimate braking distances manually while approaching corners at racing speeds.

Dynamic Guidance, Not Automated Driving

Brake Aid provides braking guidance only.

It does not apply the brakes, alter vehicle behaviour, steer the vehicle or influence throttle input.

The driver remains entirely responsible for controlling the vehicle.

Brake Aid simply provides a consistently calculated braking reference that allows the driver to focus on executing the corner rather than continually estimating braking distances.

The objective is not to replace driver skill, but to improve consistency, accelerate learning and provide a repeatable framework for refining braking performance.

Braking Datasets

The Foundation of Brake Aid

Every Brake Aid session revolves around a **braking dataset**.

A braking dataset is a complete collection of braking information for a specific combination of:

- Supported sim racing title
- Race circuit
- Vehicle

Within that dataset is the information required for Brake Aid to calculate dynamic brake markers for every significant braking zone around the circuit.

Whenever Brake Aid provides braking guidance, it does so using the currently active braking dataset.

Rather than thinking of Brake Aid as generating individual brake markers, it is more accurate to think of Brake Aid as managing braking datasets. The dynamic brake markers displayed during a race session are simply the real-time representation of the braking information contained within the active dataset.

Each user profile maintains its own collection of braking datasets, allowing different drivers to develop and refine braking references independently.

Three Ways to Obtain a Braking Dataset

When a supported game session is detected, Brake Aid determines whether a braking dataset already exists for the current game, track and vehicle combination.

Depending on the circumstances, a braking dataset may originate from one of three sources.

Importing an External Dataset (Recommended)

The recommended way to begin using Brake Aid is by importing a braking dataset collection created by an experienced Brake Aid user.

Experienced drivers can use Brake Aid's Adaptive Mode to progressively develop and refine braking datasets over multiple laps before exporting them as a shareable dataset collection.

A dataset collection may contain braking datasets covering many different games, race circuits and vehicles.

Once imported into Brake Aid, the available datasets are displayed in a hierarchical tree organised by:

- Game
- Track
- Track Configuration (where applicable)
- Vehicle

This allows you to easily determine which supported game session to start in order to use one of the available datasets.

When Brake Aid detects a running game session that matches one of the imported datasets, you may choose to begin the session using that external dataset.

The imported dataset is copied into your own user profile and becomes your active braking dataset for the session.

Brake Aid automatically starts in **Dataset Hold Mode**, allowing you to immediately begin driving using the imported braking references while preserving the integrity of the original dataset.

This workflow provides the quickest path to using Brake Aid, allowing new users to benefit immediately from braking datasets developed and refined by experienced drivers.

Resuming One of Your Own Datasets

Whenever a Brake Aid session finishes, the active braking dataset is automatically saved against your user profile for the current game, track and vehicle combination.

If you later return to the same combination, Brake Aid allows you to resume using your previously saved dataset.

This enables you to continue refining your own braking references over multiple driving sessions without needing to repeat the initial learning process.

Brake Aid will start in the same operating mode (either Dataset Hold Mode or Adaptive Mode) that it was in, at the time that the previous dataset was saved.

Creating a New Dataset

If neither an imported dataset nor one of your own saved datasets exists for the detected game, track and vehicle combination, Brake Aid can create a completely new braking dataset.

In this case, the session begins in **Adaptive Mode**.

Because no braking dataset yet exists, Brake Aid first needs to observe your braking performance around the circuit.

The first lap therefore becomes the **benchmark lap**.

During the benchmark lap, Brake Aid continuously records your braking behaviour for every significant braking zone but does not provide any dynamic brake markers.

At the completion of the lap, the recorded braking events are analysed to generate the first braking dataset for that session.

From the second lap onwards, Brake Aid is able to provide dynamic braking guidance using the newly generated dataset.

Although creating a new dataset remains an important feature of Brake Aid, it is now primarily intended for situations where no suitable braking dataset is already available.

Building and Refining Datasets

Braking datasets are not intended to be static.

As your driving improves, vehicle setups change or new braking zones emerge, the braking dataset can continue evolving alongside your driving.

Brake Aid provides two complementary methods for refining braking datasets.

In **Adaptive Mode**, the braking dataset evolves automatically as Brake Aid analyses each completed lap.

In **Dataset Hold Mode**, the dataset remains fixed while individual braking markers are manually refined using Brake Aid's offset controls.

Because the two modes can be switched at any time during a driving session, you are free to choose whichever approach best suits the current situation.

For example, you may use Dataset Hold Mode for the majority of a race weekend while relying on a proven braking dataset, then temporarily switch into Adaptive Mode to allow Brake Aid to learn braking behaviour for a previously flat-out corner that now requires braking due to increased vehicle performance.

Once the updated braking behaviour has been incorporated into the dataset, you can simply return to Dataset Hold Mode and continue driving using the revised dataset.

Sharing Braking Knowledge

One of the most powerful features of Brake Aid is the ability to share braking datasets between users.

Rather than every driver independently developing braking references from scratch, experienced Brake Aid users can create refined braking datasets and distribute them as dataset collections.

Other drivers can immediately begin using these datasets, allowing them to benefit from proven braking references while still retaining complete control over how those references are personalised.

This makes Brake Aid equally valuable as both a driver development tool and a platform for sharing braking knowledge across the sim racing community.

Adaptive Mode

Overview

Adaptive Mode is Brake Aid's learning mode.

The **circular outer ring** of the Brake Aid overlay graphic is **coloured black** when Adaptive Mode is active.

In this mode, Brake Aid continually analyses your braking performance and uses the information collected during each completed lap to generate an updated braking dataset for the current game, track and vehicle combination.

Rather than requiring the driver to manually redefine braking references whenever their driving improves, Adaptive Mode allows the braking dataset to evolve naturally alongside the driver's performance.

Adaptive Mode is typically used when:

- Creating a completely new braking dataset.
- Refining an existing dataset.
- Developing braking datasets for sharing with other Brake Aid users.
- Allowing Brake Aid to adapt to changes in driving technique, vehicle setup or racing conditions.

Although Adaptive Mode continually updates the braking dataset, the braking guidance displayed during a lap always originates from the dataset generated **before** that lap began.

This allows the current lap to be driven using a stable set of braking references while Brake Aid simultaneously observes your braking performance in the background.

Creating a New Braking Dataset

When no braking dataset exists for the current game, track and vehicle combination, Brake Aid automatically begins the session in Adaptive Mode.

Because Brake Aid has no braking information available, the first lap becomes the **benchmark lap**.

During this lap Brake Aid records every significant braking event around the circuit.

Although no dynamic brake markers are displayed during the benchmark lap, Brake Aid is continuously collecting information required to build the first braking dataset.

At the completion of the benchmark lap, the recorded braking events are analysed to generate the initial braking dataset.

From the second lap onwards, Brake Aid is then able to calculate and display dynamic brake markers for each recognised braking zone.

The benchmark lap is therefore not a calibration procedure. It is simply the first opportunity for Brake Aid to observe your braking behaviour and establish the initial braking dataset.

Continuous Dataset Development

Once a braking dataset exists, Adaptive Mode enters a continuous development cycle.

At the beginning of each lap, the most recently generated braking dataset becomes the active dataset used to calculate the dynamic brake markers displayed to the driver.

While these braking markers are guiding the current lap, Brake Aid simultaneously records a completely new set of braking events based on your actual driving performance.

Throughout the lap, Brake Aid monitors the braking characteristics of every significant braking zone, including where braking begins, how the brakes are applied, vehicle deceleration and the resulting minimum speed achieved within each corner.

At the completion of the lap, these newly recorded braking events are analysed to generate an updated braking dataset

This updated dataset then replaces the previous dataset and becomes the active braking dataset for the following lap.

The process repeats continuously for every subsequent lap, allowing the braking dataset to evolve naturally as your driving changes.

Why Adaptive Mode Updates the Entire Dataset

Brake Aid updates the braking dataset as a complete unit rather than modifying individual corners independently.

Although a single driving incident may only occur at one corner, its effects often continue throughout the remainder of the lap.

For example, braking unusually early for one corner may reduce the vehicle's exit speed.

The lower exit speed may then reduce the approach speed to the following corner, altering the braking behaviour observed there as well.

Similarly, running wide, missing an apex or encountering traffic may influence braking performance for several subsequent corners.

By analysing the complete lap as a single braking dataset, Brake Aid preserves the natural relationship between every braking zone around the circuit.

This produces braking datasets that more accurately represent how the lap was actually driven.

Refining Brake Markers

Adaptive Mode allows every dynamic brake marker to be refined using Brake Aid's Push Forward and Pull Back controls.

After completing a braking zone, the driver may adjust the braking marker for that corner in two-metre increments.

These adjustments are not applied directly to the braking marker currently being displayed.

Instead, they are applied to the braking event that Brake Aid has just recorded for the current lap.

When the updated braking dataset is generated at the end of the lap, these adjusted braking events become part of the newly generated dataset.

As a result, the driver is continually refining the next braking dataset while driving the current lap.

This allows braking performance to be improved in small, deliberate steps without requiring the driver to estimate braking distances manually while approaching corners at racing speeds.

Dynamic Learning

Adaptive Mode has been designed to accommodate the natural progression of a driver's performance.

As confidence increases, braking points may gradually move closer to the corner.

- Vehicle setups may improve.
- Fuel load may reduce.
- Tyres may change.
- Previously flat-out corners may eventually require braking as approach speeds increase.

Rather than requiring the driver to rebuild their braking references manually, Adaptive Mode allows these changes to become incorporated naturally into future braking datasets.

The result is a braking dataset that evolves alongside the driver while preserving the consistency that Brake Aid is designed to promote.

When to Use Adaptive Mode

Adaptive Mode is generally the preferred operating mode whenever the objective is to develop or improve a braking dataset.

Typical situations include:

- Learning a new circuit.
- Creating braking datasets for sharing with other Brake Aid users.
- Refining your own braking technique.
- Updating an existing braking dataset after significant improvements in driving performance.
- Incorporating new braking zones into an existing braking dataset.

Once the braking dataset reaches a level that you are satisfied with, you may choose to switch to Dataset Hold Mode to preserve the dataset while continuing to make controlled manual refinements.

Dataset Hold Mode

Overview

Dataset Hold Mode allows you to use a braking dataset as a **stable reference system** while still retaining full control over how individual braking markers are refined.

Unlike Adaptive Mode, where the braking dataset is continuously regenerated after each lap, Dataset Hold Mode preserves the active dataset indefinitely.

The dataset does not change unless explicitly modified by the driver.

This mode is primarily designed for:

- Using shared datasets created by experienced drivers.
- Practising consistent braking technique against a stable reference.
- Fine-tuning individual braking zones without altering the underlying dataset structure.
- Maintaining repeatability across multiple laps, sessions or practice sessions.

Dataset Hold Mode is the recommended operating mode when using imported or professionally prepared braking datasets.

The **circular outer ring** of the Brake Aid overlay graphic is **coloured red** when Dataset Hold Mode is active.

Core Behaviour

When Dataset Hold Mode is active, Brake Aid behaves as follows:

- The current braking dataset remains fixed and is not replaced after each lap.
- No new braking dataset is generated from driving performance.
- Dynamic brake markers continue to be calculated in real time using the active dataset.
- Brake marker positions can still be adjusted using offset inputs.
- These adjustments modify the active dataset directly rather than generating a new dataset.

This creates a stable training environment where braking guidance remains consistent unless the driver chooses to modify it.

Active Offsets and Persistent Offsets

Dataset Hold Mode introduces a two-stage offset system that allows braking markers to be refined without destabilising the dataset.

Active Offset

An **active offset** is a temporary adjustment applied to a specific braking marker based on the driver's most recent input.

When you use the **Push Forward** or **Pull Back** controls after a braking zone:

- The adjustment is stored as an active offset for that corner.
- This active offset is immediately reflected in the braking guidance on the next lap.

- The anticipation light will reflect the active offset on approach to that corner.

Active offsets represent *intent*: a change the driver has decided to apply for the next driving cycle.

Persistent Offset

Once a braking zone has been passed on the following lap:

- The active offset is converted into a persistent offset.
- The active offset is reset to zero.
- The persistent offset remains permanently applied to that braking marker for all subsequent laps.

Persistent offsets represent *established calibration* of a braking marker within the dataset.

Unlike Adaptive Mode, these offsets do not trigger further dataset regeneration.

They exist purely as adjustments applied on top of a stable dataset.

Anticipation Light Behaviour in Dataset Hold Mode

Brake Aid communicates brake markers using the on-screen anticipation light.

In Dataset Hold Mode, the behaviour of the anticipation light reflects both the underlying dataset and any active offsets.

Default State (No Active Offset)

- Light colour: Yellow
- Indicates the braking marker is unchanged from the base dataset.

Active Positive Offset (Brake Earlier)

- Light colour: Red
- Indicates the driver has requested an earlier braking point for the next lap.

Active Negative Offset (Brake Later)

- Light colour: Green
- Indicates the driver has requested a later braking point for the next lap.

Once the braking zone is passed:

- The active offset is applied to the dataset as a persistent offset.
- The active offset is cleared.
- On subsequent laps, the light returns to yellow unless a new active offset is applied.

This system ensures that only *intended changes for the next lap* affect the visual anticipation cue, while established calibration changes remain invisible but continuously applied.

Why This System Exists

Traditional driving aids often force a trade-off between:

- Stability (consistent reference points), and
- Adaptability (ability to refine braking points)

Dataset Hold Mode is designed to remove that trade-off.

By separating offsets into:

- **Active (temporary intent)**, and
- **Persistent (applied calibration)**

Brake Aid allows drivers to:

- Experiment with braking adjustments lap-to-lap,
- Without destabilising a proven dataset

This is particularly powerful when using datasets created by experienced drivers, where the goal is to preserve the integrity of the original braking structure while still allowing personal refinement.

Switching Between Adaptive and Dataset Hold Modes

Dataset Hold Mode can be entered in several ways:

- Importing a shared dataset collection (Brake Aid will default to starting in Dataset Hold Mode)
- Resuming an existing saved dataset, where Brake Aid was in Dataset Hold Mode when that dataset was saved.
- Manually switching from Adaptive Mode during a session using the **mode toggle function**, mapped to either a physical button or keyboard key.
- The **colour** of the outer ring of the Brake Aid overlay graphic turns **red** to signal Dataset Hold Mode is active.

At any time, the driver may switch back to Adaptive Mode.

When this occurs:

- The **colour** of the outer ring of the Brake Aid overlay graphic turns **black** to signal Adaptive Mode is active.
- The current dataset remains active,
- But Brake Aid resumes generating new braking data from observed driving performance,
- Allowing the dataset to evolve again.

This flexibility allows a single session to combine:

- Stable reference driving (Dataset Hold Mode), and
- Learning and expansion (Adaptive Mode)

without requiring session restarts.

Summary

Dataset Hold Mode transforms Brake Aid from a learning system into a **precision refinement system**.

It allows:

- Stable use of expert-developed braking datasets,
- Controlled per-corner adjustment,
- Persistent calibration of braking markers,
- And full interoperability with Adaptive Mode when required.

It is the primary mode intended for structured driving, consistent practice, and use of shared braking datasets.

Shared Dataset Collections

Overview

Shared Dataset Collections allow braking datasets created within Brake Aid to be exported, distributed and imported between users.

This transforms Brake Aid from a purely individual training tool into a system for sharing braking knowledge across the sim racing community.

Instead of every driver independently learning every braking zone from scratch, experienced users can develop high-quality braking datasets and share them with others in a structured, reusable format.

What a Dataset Collection Contains

A dataset collection is a packaged set of braking datasets grouped by:

- Game
- Track
- Vehicle

Each dataset within the collection represents a complete braking reference system for that specific combination.

A single collection may contain multiple datasets covering different circuits and vehicles, allowing a driver to build or import an entire library of braking references.

When imported into Brake Aid, these datasets become available for immediate use during supported game sessions.

Creating a Dataset Collection (Exporting)

Experienced Brake Aid users can create dataset collections by exporting braking datasets that have been developed during Adaptive Mode sessions.

A typical workflow for creating a dataset collection is:

1. Start a sim racing session in Adaptive Mode.
2. Drive multiple laps to allow Brake Aid to generate and refine a braking dataset.
3. Continue refining the dataset until it reflects the desired level of braking performance and consistency.
4. Exit the sim racing session which will prompt Brake Aid to save the latest braking dataset from the session to your user profile.
5. Navigate to the **User Selection** Tab in the **Brake Aid Setup** Page of the UI.
6. Select your user from the users list
7. Click **Export Your Brake Datasets**.
8. Save your collection of braking datasets to a Brake Aid Dataset Collection (.bds) file to share with other users.

Each exported dataset reflects the final state of that braking dataset at the time of export, including all refinements made during Adaptive Mode and any manual adjustments applied during Dataset Hold Mode.

This ensures that shared datasets are consistent, repeatable and ready for immediate use by other drivers.

Importing Dataset Collections

Imported dataset collections allow users to instantly access braking datasets created by other Brake Aid users.

To import a Brake Aid dataset collection (.bds) file:

- Click **LOAD DATASETS** on the left-hand side of the Brake Aid UI.
- Click **Load External Dataset Collection**
- Open a **.bds file** that has been shared with you.

When a dataset collection is imported, Brake Aid:

- Reads all available datasets in the collection.
- Organises them into a hierarchical structure:
 - **Game**
 - **Track**
 - **Track Configuration**
 - **Vehicle**
- Displays the available datasets in a selection interface.

This allows the user to quickly identify which game session must be started in order to use a specific dataset.

Using an Imported Dataset

When a compatible game session is detected, and a matching dataset exists within an imported collection, Brake Aid provides a third session option:

- Start New Session in Adaptive Mode (No Dataset)
- Continue With Your Previous Dataset
- **Start Session with External Dataset**

When this option is selected:

- The dataset is copied into the user's active profile.
- Brake Aid begins the session in Dataset Hold Mode.
- The imported dataset remains unchanged in its original form.
- All modifications are applied only to the user's local copy.

This ensures that shared datasets remain intact while still allowing full personalisation for each user.

Why Shared Datasets Matter

Shared Dataset Collections fundamentally change how Brake Aid is used.

Instead of requiring every driver to:

- learn every circuit independently,
- build braking references from scratch,
- and refine each braking zone over multiple sessions,

Brake Aid allows drivers to immediately begin using braking datasets that have already been developed by experienced users.

This provides:

- Faster onboarding for new drivers
- More consistent braking guidance across users
- A structured way to distribute driving expertise
- A scalable model for community-driven improvement

Experienced drivers effectively become dataset creators, while newer drivers become dataset users who progressively refine those references to match their own driving style.

Relationship to Adaptive and Dataset Hold Modes

Shared Dataset Collections are designed to work seamlessly with both operating modes.

- In Dataset Hold Mode, imported datasets are used as stable references for consistent driving and manual refinement.
- In Adaptive Mode, datasets can be further developed based on the user's own driving performance.

This allows a full lifecycle:

1. Expert driver creates dataset (Adaptive Mode)
2. Dataset is exported and shared
3. Another user imports dataset (Dataset Hold Mode)
4. User refines dataset for their own style
5. Optional return to Adaptive Mode for further development

This cycle enables braking knowledge to evolve across both individuals and the wider community.

Summary

Shared Dataset Collections allow Brake Aid to function not only as a driver aid, but as a distribution system for braking expertise.

They enable:

- Exporting complete braking datasets
- Sharing expert-developed braking references
- Immediate onboarding for new users

- Structured dataset organisation by game, track and vehicle
- Seamless integration with Adaptive and Dataset Hold Modes

This creates a scalable ecosystem where braking knowledge can be developed once and reused many times, while still allowing every driver to personalise their own experience.

Customising Brake Markers (Offsets System)

Overview

Brake Aid allows individual braking markers to be fine-tuned using a per-corner offset system.

While braking datasets provide the base structure for all braking guidance, offsets allow each braking zone to be adjusted to better suit the driver's personal braking style, confidence level, or evolving performance.

Offsets can be applied in two ways depending on the active operating mode:

- Adaptive Mode (offsets influence dataset generation)
- Dataset Hold Mode (offsets modify the active dataset directly)

Although the internal behaviour differs between modes, the user interaction remains identical.

Offset Controls

Brake Aid provides two primary inputs for adjusting braking markers:

- Push Forward → moves the braking marker closer to the corner (brake later)
- Pull Back → moves the braking marker further from the corner (brake earlier)

These inputs can be mapped to either physical buttons on an external peripheral, or keyboard keys from within the **BUTTON MAPPING** page of the Brake Aid UI.

Each input adjusts the selected braking marker in fixed increments of 2 metres* per press.

**Increment distance is customisable within Brake Aid's CONFIGURATION settings page.*

The inputs can be pressed multiple times to achieve larger offsets.

For example, 3 presses of the Push Forward input will apply a 6-metre negative offset when the individual increment value is set at the default 2 metres.

The Brake Aid overlay light graphic will **briefly flash** either **red** (for Pull Back) or **green** (for Push Forward) each time an offset input has been pressed to let the user know the offset has been successfully applied.

Offset Behaviour in Adaptive Mode

In Adaptive Mode, offset inputs are applied to the recorded braking event, not the current displayed marker.

This means:

- The current lap is driven using the existing braking dataset.
- Offset inputs are stored against the braking event being observed.
- At the end of the lap, a new braking dataset is generated.
- The applied offsets influence the next dataset generation.

This allows drivers to refine braking behaviour while driving, without altering the stability of the current lap's guidance.

Offset Behaviour in Dataset Hold Mode

In Dataset Hold Mode, offset behaviour is fundamentally different.

Instead of influencing future dataset generation, offsets directly modify the active dataset.

This introduces a two-stage system:

Active Offset

A temporary adjustment applied after the driver completes a braking zone.

- Stored immediately after input
- Reflected in the next approach to that corner
- Used to update the braking marker for the next lap

Persistent Offset

Once the corner has been driven again:

- The active offset is converted into a persistent offset
- Persistent offsets are permanently applied to that braking marker
- Active offset is reset to zero

This allows drivers to make deliberate adjustments lap-to-lap while maintaining a stable dataset structure.

Per-Corner Refinement Philosophy

Offsets are designed to support incremental improvement rather than large adjustments.

The default 2-metre resolution ensures that:

- Changes remain precise
- Adjustments are deliberate
- Braking development remains gradual and controlled

This is particularly important when using imported datasets, where the goal is typically to refine rather than rebuild braking behaviour.

The Anticipation Light

Overview

Brake Aid communicates all braking markers using the **anticipation light system**, displayed as an on-screen overlay indicator.

The anticipation light is the primary visual cue used to indicate when braking should begin.

It is designed to provide both:

- Early preparation for braking
- Precise timing for brake initiation

Basic Operation

In its simplest form, the anticipation light operates as follows:

- The light illuminates approximately one second* before the calculated braking point is reached.
**Anticipation light time is customisable within Brake Aid's CONFIGURATION settings page.*
- The light remains active during the final approach phase.
- The light turns off at the exact moment the braking marker is reached.

This transition from ON to OFF represents the precise braking trigger point.

Default Colour Behaviour

When no offsets are applied, the anticipation light displays

Yellow → Default braking marker state

This indicates that the braking marker is derived directly from the active dataset without any additional modifications.

Offset-Induced Colour Changes

When offsets are applied, the anticipation light reflects the direction of the adjustment.

Positive Offset (Brake Earlier)

- Colour: Red
- Meaning: braking marker has been moved further away from the corner
- Interpretation: driver is expected to brake earlier than the base dataset

Negative Offset (Brake Later)

- Colour: Green
- Meaning: braking marker has been moved closer towards the corner
- Interpretation: driver is expected to brake later than the base dataset

These colours provide immediate feedback before reaching the braking zone, reinforcing the driver's most recent adjustment.

Behaviour in Dataset Hold Mode

Dataset Hold Mode introduces a separation between:

- Active Offset (temporary intent)
- Persistent Offset (applied calibration)

This affects how the anticipation light behaves.

Active Offset Phase

When a driver applies an offset:

- This indicates a pending adjustment for the next approach
- The anticipation light reflects the active offset colour (red or green) on the next approach to that corner

After Corner Completion

Once the driver has passed the braking zone:

- The active offset is converted into a persistent offset
- The active offset is cleared
- The anticipation light returns to yellow for that corner on subsequent laps (unless a new active offset is applied)

Persistent Offset Behaviour

Persistent offsets:

- Do not affect the colour of the anticipation light
- Are silently applied to all future braking calculations
- Represent the established braking calibration for that corner

This ensures that only intentional, not historical, adjustments are visually communicated.

Design Purpose

The anticipation light is designed to eliminate ambiguity in braking timing.

Rather than requiring the driver to interpret visual trackside references at high speed, Brake Aid provides a clear binary cue:

- Light ON → prepare to brake
- Light OFF → brake now

Offset colouring adds a secondary layer of information, allowing the driver to understand whether braking behaviour has been recently adjusted.

Summary

The combination of the offset system and anticipation light creates a unified feedback loop:

- Brake Aid observes braking behaviour
- Driver adjusts individual corners
- System reflects those changes visually in real time
- Adjustments are stabilised through persistent offsets (Dataset Hold Mode)

This ensures that braking refinement remains both controlled and immediately understandable, even at racing speed.

Don't Count Lap Function

Overview

The Don't Count Lap function allows the driver to exclude the current lap from being used to update or influence the braking dataset **in Adaptive Mode**.

This feature is designed to protect braking datasets from being affected by laps that do not accurately represent normal driving conditions.

The Don't Count Lap function input is **mapped by the user** to either a physical button on an external peripheral, or a keyboard key from within the **BUTTON MAPPING** page of the Brake Aid UI.

When This Feature Is Used

A lap may become unsuitable for dataset generation when external factors affect braking performance, such as:

- Traffic or obstruction from other vehicles
- Loss of control or track excursions
- Unusual or experimental driving lines
- Missed braking points due to error
- Any situation where braking behaviour is not representative of intended performance

In these cases, allowing the lap to update the braking dataset may degrade future braking guidance.

Behaviour in Adaptive Mode

When Don't Count Lap is activated during Adaptive Mode:

- The current lap is flagged as invalid for dataset generation.
- Brake Aid continues to record braking events during the lap.
- However, these events are discarded at the end of the lap.
- The previous braking dataset remains unchanged.
- The next lap continues using the same dataset as the current lap.

This ensures that only representative laps contribute to dataset evolution.

Behaviour in Dataset Hold Mode

In Dataset Hold Mode:

- The Don't Count Lap function input has no effect while Brake Aid is operating in Dataset Hold Mode.

Visual Feedback

When Don't Count Lap is activated:

- The anticipation light flashes in a rapid sequence:
- Red → Yellow → Green → Off
- This confirms that the current lap has been marked as excluded from dataset updates.

Purpose

The purpose of this feature is to ensure dataset integrity.

Brake Aid relies on consistent braking behaviour to generate meaningful braking datasets. A single irregular lap can negatively influence braking markers for multiple corners.

Don't Count Lap provides a simple mechanism to preserve dataset quality without interrupting the driving session.

Using the Software

Launching Brake Aid

In the Brake Aid Beta root directory, launch ***Brake Aid Beta.exe*** to open the Brake Aid user interface.

Brake Aid Setup Menu

The **Brake Aid Setup** menu allows the user to create or select a user profile, and then to start a Brake Aid *session* once a compatible game session has been found.

User Selection Tab

Creating a user profile:

To Create a new user profile, click the *Create New User* button, then type your name in the popup text box, then click the *OK* button.

Selecting a user profile

Click your profile name from the *users list*.

On the right-hand side, the latest reaction time the system has on record for the selected user profile is displayed.

(The default reaction time for new user is 200ms, although this is updated dynamically when the new user actively uses Brake Aid)

After selecting your user profile, click *NEXT* at the bottom right of the window.

Begin Session Tab

Detecting a Game Session

When the Begin Session Tab is displayed:

- The application scans for supported sim racing titles.
- It identifies active game sessions.
- It determines the current:
 - Game
 - Track
 - Vehicle

This information is used to match against available braking datasets.

Session Start Options

Once a valid session is detected, Brake Aid provides the following options:

1. Start Session with External Dataset

This option is available when a matching dataset exists in an imported dataset collection.

When selected:

- The external dataset is copied into the user profile.
- Brake Aid starts in Dataset Hold Mode.
- The dataset is immediately active.
- No benchmark lap is required.

This is the recommended workflow for most users.

2. Continue With Your Previous Dataset

If the user has previously used Brake Aid with the same game, track and vehicle combination:

- The last saved dataset is loaded.
- Brake Aid resumes in the mode it was last used in.
- Any previously applied persistent offsets remain active.

This allows continuity across sessions without reinitialization.

3. Start New Session in Adaptive Mode (No Dataset)

If no suitable dataset exists, or if the user chooses to begin fresh:

- Brake Aid starts in Adaptive Mode.
- The first lap becomes the benchmark lap.
- No braking guidance is provided during this lap.
- A new braking dataset is generated at the end of the lap.

This mode is primarily used for creating new datasets or rebuilding existing ones.

Once a Brake Aid session is launched, the UI window will **minimise to the system tray**.

To recall the UI during a session, right click the *Brake Aid icon* in the system tray and select *Show UI*.

Configuring the Brake Aid Overlay Light Graphic

Important: In order for the Brake Aid overlay graphic to be displayed over the game screen, as with most other third-party in-game telemetry overlays, the game itself must be run in either *Windowed Mode*, or *Full Screen Borderless Window Mode*.

In iRacing, *Full Screen Borderless Window Mode* is enabled from within a running iRacing sim session, by heading to:

Settings → Display Settings → Display,
then toggling *Full Screen* to Off, and *Border* to Off.

(The sim session may need to be restarted for the changes to take effect.)

Once a Brake Aid session is started/resumed, the overlay graphic will be displayed when the racing session itself is started (i.e. the game enters *driving mode*).

Once the overlay is displayed, it can be freely dragged anywhere on the screen and can be resized by either *dragging from the bottom right corner* of the overlay, or by *scrolling the mouse wheel* while the cursor is positioned over the overlay.

Exiting a Brake Aid Session

Once a Brake Aid session has begun, it will *automatically* exit (and save the user's latest braking dataset to their user profile) once the game session it is attached to *completely* ends.

Alternatively, the session can be ended at any time from the Brake Aid UI (which can be displayed by right clicking the Brake Aid icon in the system tray and clicking *Show UI* and then clicking the *End Session* button in the *Begin Session* tab of *Brake Aid Setup*).

Button Mapping

Brake Aid currently has four user inputs, which include:

- Push Forward
- Pull Back
- Don't Count
- Adaptive/Dataset Hold Mode Toggle

Each of these functions can be mapped to either a keyboard key, or a button on a connected external peripheral, such as game controllers, racing sim wheels and other sim hardware that contains buttons.

Game Controller Compatibility

Most standard game devices that work on Windows should be compatible with Brake Aid.

Devices that need third-party drivers installed to operate correctly, may also be compatible with Brake Aid if they are listed within the Windows *Game Controllers* control panel applet, and the buttons are responsive within the *Test* tab of that device's *properties* menu, however this is ***not guaranteed***.

Regarding Keyboard Key Binding:

In addition to binding Brake Aid's user inputs to external peripheral buttons, alternatively, users also have the option to bind any of these inputs to keyboard keys.

Brake Aid assigns mapped inputs to keyboard keys by allocating each mapped key as a *hotkey* when a Brake Aid session is started/resumed.

During a Brake Aid session, any mapped keyboard keys are effectively *blocked* from the rest of the PC and are *only* operational with Brake Aid during this time.

When a Brake Aid session is ended, the mapped keyboard keys are deallocated from their hotkey state, and they become responsive to the rest of the PC again.

Only during an *active* Brake Aid session, are mapped keyboard keys blocked from the rest of the PC.

Mapping a Button/Key to a Brake Aid User Input Function:

On the left-hand side of the Brake Aid user interface, click *Button Mapping*.

To map one of the user input functions to a button or key, click the *MAP* button in that function's row.

If successful, the text under *Current Mapping* will be updated to confirm that the mapping was successful.

To remove a user input mapping, click *CLEAR MAPPING* in that function's row.

Brake Aid Configuration

The *configuration* page within the Brake Aid user interface contains various runtime variables that alter the behaviour of Brake Aid.

The default settings are optimised to allow Brake Aid to work effectively on a variety of racing setups.

If Brake Aid isn't working correctly or effectively for you, please reach out to the support team, who can assist with tweaking the configuration options to better suit your particular platform.

Feel free to adjust the Brake Aid configuration parameters if you want to experiment.

If you want to roll back to the default configuration options, click *Load Defaults* at the bottom right corner of the configuration page.

Troubleshooting

As Brake Aid is currently beta software, there is a high likelihood that operational issues may arise, and bugs may be discovered.

If the software crashes

The beta version of the Brake Aid software has extensive internal logging built in, which logs technical warnings and errors encountered in each of the subsystems of the software as it is running.

This beta version saves the complete set of logs for *up to 10 consecutive runs* of the Brake Aid software.

The most effective way for the support team to assist you with software crashes is to package the complete set of logs and email them to support.

This is easily achieved by double clicking the batch script ***Dump Brake Aid Logs to zip folder.bat*** which is located within the root folder of the Brake Aid software.

Once this script runs, a file called ***LogOutput.zip*** appears in the root folder. It is this zip file that should be sent to support, along with a brief description of the circumstances surrounding the software crash.

Reporting Bugs

There are likely to be many bugs and glitches that occur with this beta version of the Brake Aid software!

When a bug or glitch is encountered, please make a note of it and post it on the Brake Aid Beta Testing Discord page.

Third Party Notices

This software uses third-party components.

For acknowledgements and license information, see *Third Party Licenses/ThirdPartyNotices.txt* included with this software.